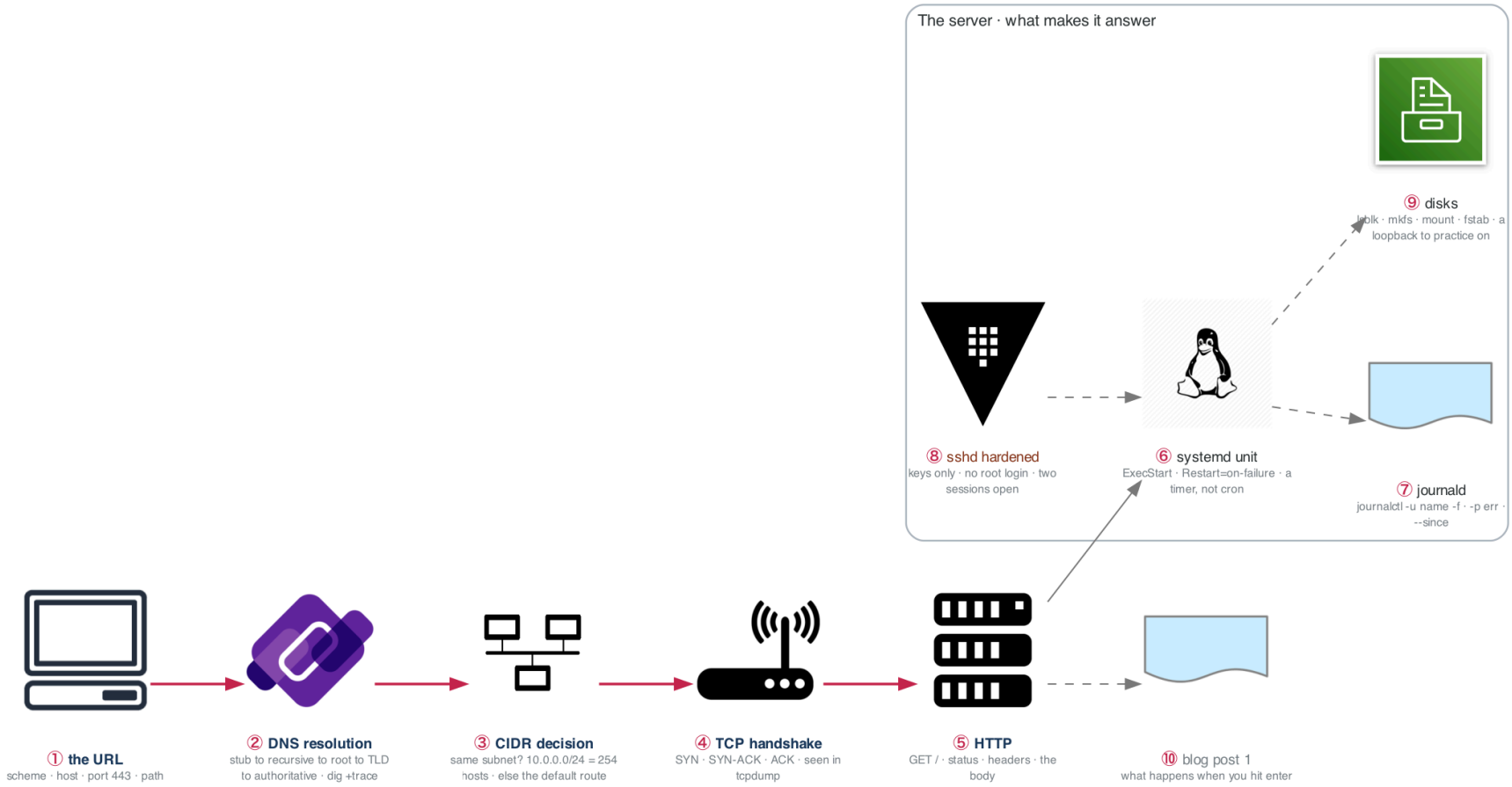
From a URL to a packet, and the service that answers, numbered

Week 2 · Figure 2.1 · figure with badges 1–10 · 2026-10-17



What to say, badge by badge

About three and a half minutes at a normal pace (403 words). Point at each badge as you say its line.

▶ **Open**
“This is the week's deliverable drawn out: what happens when you type a URL and hit enter, and what has to be true on the server for anything to come back. Left to right is the request. The box is the server.”

1 **the URL**
“Four parts: a scheme that implies a port, a host that needs resolving, a path the server interprets. 443 is a number you say without thinking by Friday.”

2 **DNS resolution**
“Your stub resolver asks a recursive resolver. It walks root, then the TLD, then the authoritative server, caching each answer for its TTL. `dig +trace` shows every hop. A wrong answer is almost always a stale cache or the wrong authoritative server.”

3 **CIDR decision**
“Before a packet leaves, the kernel asks one question: is the destination in my subnet? A /24 is 256 addresses, 254 usable, network and broadcast reserved. If it is not local, the packet goes to the default route. One hundred problems this week make that question automatic.”

4 **TCP handshake**
“SYN, SYN-ACK, ACK. Three packets, one round trip, before a single byte of HTTP. In tcpdump you can point at all three. A connection that hangs at SYN is a firewall or a routing problem, never the application.”

5 **HTTP**
“Now the request: a method, a path, headers, a body. The status code is the server's one-word answer. Everything above this line was plumbing; everything the user sees starts here.”

6 **systemd unit**
“The server answers because a unit file says ExecStart and Restart=on-failure, and systemd keeps it running. A timer unit replaces cron with logging, dependencies, and `systemctl list-timers`. Write both from blank.”

7 **journald**
“`journalctl -u name -f` follows one service. `-p err --since today` filters. This is where you look before you look anywhere else.”

8 **sshd hardened**
“Keys only, no root login, no passwords. The rule that matters: keep two sessions open while you change sshd, and test from a third before you close anything. Lockouts are the week's classic self-inflicted incident.”

9 **disks**
“lsblk, mkfs, mount, fstab. Practice on a loopback file so a mistake costs nothing. A full disk is BF-01; knowing the layout is how you find what filled it.”

10 **blog post**
“Saturday you write all of this down for a stranger. If a step is hard to explain, that is the step you do not yet own.”

■ **Close**
“Name, route, handshake, request. Then a unit that keeps the service alive and logs you can read. Every VPC and every Kubernetes Service later is this picture with different names on the boxes.”